

EXPRESSION OF INTEREST

PROJECT BRIDGE

National Digital Economy University-Led Research Clusters Programme

Cluster E: Trust, Safety, Consumer Protection, and Online Harms

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1 Introduction and Expression of Interest

Bayero University, Kano (BUK) leads this consortium in submitting an Expression of Interest (EOI) for Cluster E: Trust, Safety, Consumer Protection, and Online Harms.

We respond to the Federal Ministry of Communications, Innovation and Digital Economy (FMCIDE) call (Reference No: NG-FMCIDE-544880-CS-QCBS, Project Number P508383) with a team spanning six Nigerian and two international institutions. Our team have track records in digital trust, hate speech detection, cybersecurity, data privacy, consumer protection, Natural Language Processing for African languages, and AI governance.

This consortium combines three distinct elements: university research across six institutions, operational industry platforms (VigilSentinel for cybersecurity, Wazobia Invoice for fintech, SPIDVerify for identity verification), and hands-on government policy experience, including former Vice chancellor, former Commissioner for Science and Technology in Kano State and national leadership of the UBEC Digital Resource Center. This academia–industry–government model is what has historically produced research that changes things, from MIT’s Kendall Square to the Cambridge tech cluster.

Timing makes this urgent. As Nigeria lays at least 90,000 km of fibre-optic backbone under Project BRIDGE, millions of new users will come online. Many will meet digital services, digital markets, and digital risks for the first time. Fibre alone doesn’t create trust. Our research will ensure that infrastructure and knowledge grow in step, so expanded connectivity arrives with the governance tools, safety mechanisms, and institutional muscle to protect citizens and sustain confidence in Nigeria’s digital future.

2 Consortium Structure and Participating Institutions

Six institutions form this Research Cluster: three Nigerian and two UK-based international partners. This composition meets the Programme’s four-to-six institution requirement with the encouraged inclusion of international partners.

The consortium comprises six confirmed institutions. BUK leads on digital trust, NLP, and online safety; KUST Wudil contributes AI and machine learning; FUT Babura brings recommender systems and data analytics; Northwest University Kano adds deep learning, computer vision, and cybersecurity. Two UK international partners add cybersecurity engineering and privacy-preserving identity management.

3 Thematic Alignment and Relevance to Cluster E

Our team’s work maps tightly onto every priority in the Cluster E Terms of Reference.

S/N	Institution	Country	Role	Thematic Focus
1	Bayero University, Kano	Nigeria	Lead Institution	Online Harms, NLP, AI Safety, Hate Speech Detection
2	Kano University of Science and Technology, Wudil	Nigeria	Nigerian Partner (AI Lead)	AI, Machine Learning, Data-Driven Systems
3	Federal University of Technology, Babura	Nigeria	Nigerian Partner	Recommender Systems, Data Analytics, Cybersecurity, ICT
4	Northwest University, Kano	Nigeria	Nigerian Partner	Deep Learning, Feature Selection, Computer Vision, Cybersecurity
5	York St John University	United Kingdom	International Partner	Cybersecurity, Trust Systems, Data Privacy, IoT Security
6	University of Leicester	United Kingdom	International Partner	Cyber Security, Privacy, SSI, Forensic Investigation

Table 1: Consortium composition and thematic responsibilities.

3.1 Online Harms, Hate Speech, and Content Moderation

No other team on the continent has built what this consortium has in online harms research for African languages. Our consortium works extensively on this:

- **AfriHate (NAACL 2025):** The first large-scale hate speech and abusive language dataset spanning 15 African languages. Native speakers annotated every instance with socio-cultural context. This fills a gap that left content moderation tools blind to African-language abuse for years [1].
- **AfriSenti and NaijaSenti:** A sentiment analysis benchmark covering 14 African languages with over 110,000 annotated tweets—used worldwide to study polarisation and harmful content dynamics [3].
- **NaijaSenti:** The first large-scale sentiment dataset for Nigeria’s four most spoken languages—Hausa, Igbo, Nigerian Pidgin, and Yorùbá—for tracking online discourse patterns specific to Nigeria.
- **HERDPhobia:** A focused dataset for detecting hate speech against Fulani herdsmen—one of Nigeria’s most socially charged forms of targeted online abuse.
- **POLARIZATION (2025):** Ongoing work on how digital platforms amplify political division in African democracies and what that means for social cohesion.

3.2 Trust, Cybersecurity, and Data Privacy

Trust doesn't emerge by accident. It takes deliberate engineering. Our consortium works on this from several angles:

- **Patent-pending FinTech Security:** A zero-trust offline point-of-sale network using single permanent QR code, manager-gated one-time codes, dual-key encryption, and 12-word recovery phrase. This research produced Wazobia Invoice (www.wazobia-invoice.ng).
- **Integrated FinTech Cybersecurity:** Research on securing the FinTech stack through an integrated platform for vulnerability management and financial cyber operations, leading to VigilSentinel (www.vigilsentinel.com).
- **Trust-Based Protocols:** The Dynamic Attribute Trust Model (DATM) for secure collaborative routing in wireless networks [4, 5]. This work underpinned a completed doctoral programme.
- **Privacy by Design:** Privacy-preserving frameworks for biometric IoT systems in healthcare [7], embedding data protection into system architecture from the ground up. This research underpinned a second completed doctoral programme.
- **Voice Biometrics and IoT Security:** Research on trust and voice biometrics authentication for IoT devices [6]. This work supported the completion of two postgraduate programmes and won Best Paper at ISIC 2025, Lübeck, Germany.
- **Cross-Cultural AI Trust:** How trust in AI varies across cultures—work that shapes governance frameworks sensitive to local realities.
- **Cyber Risk and Secure Infrastructure:** Research on cyber risk management and secure 5G/6G communications, including congestion control for integrated satellite-terrestrial networks. As Project BRIDGE expands Nigeria's digital backbone, this becomes essential [11].
- **Self-Sovereign Identity:** Research on SSI, reputation frameworks, decentralised identity verification, and GDPR-compliant health data platforms [12]. This research produced SPIDVerify (spidverify.com), now serving 500+ organisations globally.
- **Deep Learning for Cybersecurity:** Research on phishing detection using focal loss and self-attention CNNs [23] and network intrusion detection using hybrid SWIN Transformer and VGG19 [24], published at IEEE CyberComp 2024—directly applicable to online fraud prevention and consumer protection.

3.3 Governance, Regulation, and Policy

Our team has experience in governance and policy enegagement.

- Prof. Galadanci served as Special Advisor to the Governor of Kano State on ICT and Education, then as Commissioner for Science and Technology. He knows what

it takes to turn research into government policy at the executive level.

- Dr. Usman directs the Cybersecurity Research Group at York St John University. His team works on regulatory compliance (GDPR, PCI DSS) and adaptive governance for emerging technologies.
- Assoc. Prof. Haruna serves as Director of the Centre for Information Technology at FUT Babura, bringing ICT governance, infrastructure management, and policy implementation experience at the institutional level. His certification in Good Governance and Leadership (National Institute for Policy and Strategic Studies, KURU) extends this into national policy work.
- Prof. Abdulrahman serves as Dean of the Faculty of Computing and Mathematical Sciences at KUST Wudil, providing institutional leadership and policy experience at the level of curriculum design, programme development, and research governance.
- Dr. Muhammad holds dual appointments as Google DeepMind Academic Fellow and Advanced Research Fellow at Imperial College London. Both roles put him at the centre of global AI governance debates.
- Across the consortium, we maintain relationships with government agencies, development partners (World Bank, Google, Lacuna Fund), and civil society organisations in Nigeria and beyond.

4 Key Personnel (Indicative)

4.1 Consortium Lead: Dr. Shamsuddeen Hassan Muhammad

Dr. Muhammad holds a PhD in Computer Science from the University of Porto (thesis: *Domain-Specific and Context-Aware Approaches to Sentiment Analysis*) and an MSc from the University of Manchester, UK. He is Advanced Research Fellow and Google DeepMind Academic Fellow at Imperial College London, and Senior Lecturer at Bayero University, Kano. Research impact: H-index 22, 4,716 citations.

Selected achievements:

- **Grants:** PI/Co-PI on over USD 1.5M+ in competitive funding, including: Wellcome Trust Co-PI (£3M, generative AI for mental health); OpenAI Co-PI (USD 50,000, multilingual AI safety); Open Data Platform PI (USD 430,000); Lacuna Fund hate speech datasets (USD 147,000); Oracle OCI Award (USD 50,000); Nigerian AI Research Grant (₦5,000,000).
- **Datasets and benchmarks:** PI of AfriHate (NAACL 2025)—hate speech dataset for 15 African languages; PI of BRIGHTER—emotion recognition dataset for 28 languages (ACL 2025 Best Resource Paper); co-developer of AfriSenti, NaijaSenti, HERDPhobia, and IrokoBench.

- **Awards:** ACL 2025 Best Resource Paper; SemEval 2025 Best Task Paper; NAACL 2025 Outstanding Paper; Deep Learning Indaba 2025 Best Paper; NeurIPS 2024 (C3NLP) Best Paper; AfricaNLP 2023 Best Paper; IJCNLP–AAACL 2023 Area Chair Award; EMNLP 2020 Wikimedia Research Award.
- **Service:** Area Chair at ACL, EMNLP, NAACL; SemEval Shared Task Organiser 2023–2026; Senate Member, BUK; Grant Review Committee Member, Imperial College London.
- **Supervision:** Three active PhD students at BUK (speech recognition, multimodal emotion recognition, machine-generated content detection). Visiting Lecturer at AIMS South Africa and AIMS Cameroon.
- Co-founder of HausaNLP and Arewa Data Science Academy.

4.2 Senior Research Specialist: Dr. Aminu Bello Usman

Dr. Usman is Associate Professor of Computer Science and Director of the Cybersecurity Research Group at York St John University. He previously served as Head of the School of Computer Science at the University of Sunderland and Head of the Department of Computer and Data Science at York St John University. He earned his PhD in Cyber Security from Auckland University of Technology and holds Senior Fellow (D3) status with Advance HE.

What he brings to Cluster E:

- Principal Investigator of *Cyber360: An Integrated Experiential Platform for Cyber Hygiene, Network Defence, and Embedded Systems Security Education* (£120,000). The project produced open-source materials on Embedded System Security, Cyber Hygiene, and Network Security.
- Recipient of ISIC 2025 Best Paper Award.
- Supervised and successfully graduated doctoral candidates in trust-based and voice biometric authentication, privacy-by-design frameworks for biometric verification, and secure authentication for financial applications—research areas at the heart of Cluster E.
- Patent pending: Zero-Trust Offline Point-of-Sale Network Using Single Permanent QR Code, Manager-Gated One-Time Codes, Dual-Key Encryption, and 12-Word Recovery Phrase. The research produced Wazobia Invoice.
- Founder of WaZoBia Smartech Technologies, comprising: WaZoBia Invoice (secure offline fintech for small enterprises), WaZoBia Books (accounting software for Nigerian businesses), and VigilSentinel (integrated cybersecurity operations platform).
- Co-founder of Kanotec Solutions, a UK-based research and technology company.

- Research on the Dynamic Attribute Trust Model (DATM), Privacy by Design frameworks for biometric IoT in healthcare, and cross-cultural trust in AI—informing governance frameworks sensitive to local contexts.

4.3 Senior Research Specialist: Dr. Idris Abdulmumin

Dr. Abdulmumin is a Postdoctoral Fellow at the University of Pretoria’s Data Science for Social Impact group and a Lecturer at Ahmadu Bello University, Zaria. His PhD (Bayero University, Kano) focused on NLP and machine translation. He brings 7+ years of hands-on experience in hate speech analysis and sentiment research.

What he brings to Cluster E:

- Co-authored AfriHate, HERDPhobia, and AfriSenti—three of the most significant hate speech and sentiment datasets for African languages.
- Studied COVID-19 vaccination misinformation in Nigerian cyberspace, showing how to track health-related disinformation at scale.
- Built ZASCA-Sum (legal document summarisation), with applications to consumer protection and regulatory analysis.
- Creates NLP tools for low-resource African languages—the building blocks for content moderation that works in local contexts.
- Active in Masakhane, HausaNLP, and the Open Language Data Initiative.

4.4 Senior Research Specialist (NLP and Ethical AI): Dr. Ibrahim Said Ahmad

Dr. Ibrahim Said Ahmad is a Lecturer in the Department of Computer Science at Bayero University, Kano, with over 10 years of research and teaching experience across NLP, ethical AI, and multilingual machine learning. He holds a PhD from Universiti Kebangsaan Malaysia and an MSc from the University of Nottingham Malaysia. His research record spans 40+ peer-reviewed publications, 10+ as first author, across NLP, emotion detection, sentiment analysis, and responsible AI.

What makes Dr. Ahmad particularly valuable to this cluster is the combination of his technical depth in low-resource African language NLP and his focus on the cultural and ethical dimensions of AI. He doesn’t just build models—he asks whether those models work fairly and accurately for African users, in African contexts.

What he brings to Cluster E:

- Co-author and active contributor to AfriHate—the first labelled hate speech dataset for 18 African languages—and to NaijaSenti, the first large-scale multilingual sentiment corpus for Nigerian languages (Hausa, Igbo, Nigerian Pidgin, Yorùbá).

- Organised SemEval 2025 Task 11: Bridging the Gap in Text-Based Emotion Detection, a major international shared task on African language emotion analysis attracting global research participation.
- Organised SemEval 2023 Task 12: Sentiment Analysis for African Languages (AfriSenti-SemEval)—the competition that produced the AfriSenti benchmark now used worldwide.
- Research on cultural alignment in LLMs: studied whether ChatGPT responses align with native Hausa speakers' cultural expectations, using 37 culturally validated prompts and psycholinguistic expert verification. Findings show significant cultural misalignment—directly relevant to Cluster E's work on consumer protection and platform accountability.
- Developed AfriEmo, a large emotion dataset for 18 African languages, supporting multi-label emotion detection, cross-lingual emotion analysis, and emotion intensity classification.
- Research on COVID-19 vaccination misinformation sentiment in Nigerian cyberspace, combining classical machine learning and fine-tuned large language models (BERT, AfroXLMR, XLM-RoBERTa).
- Supervised MSc researchers in fake news detection, vulnerability exploitation prediction, and negative purchase intent classification—research themes with direct applications to online harm detection and consumer protection.
- Won over USD 300,000 in competitive grants as PI and Co-PI, including two Google Research exploreCSR awards, two TensorFlow awards, Lacuna Fund grants totalling USD 168,000+, an Oracle OCI Award, and a Google COVID-19 Research Grant.
- Received the Best Paper Award at AfricaNLP 2023 (Kigali), Best Paper Honorable Mention at ACII 2024, and recognition in the IRCAI Global Top 100 AI solutions for the SDGs.
- Member of the Programme Committee for the 31st International Conference on Computational Linguistics (COLING 2025). Reviewer for IEEE Access, Information Processing & Management, Neural Computing and Applications, ACL, and LREC.
- Fluent in English and Hausa; working proficiency in Malay and Arabic; linguistic analysis expertise across 20 low-resource languages.

Associate Professor Khalid Haruna is Director of the Centre for Information Technology (CIT) and Head of Computer Science at the Federal University of Technology, Babura (FUT Babura), Jigawa State. He holds a PhD in Computer Science from the University of Malaya, Malaysia, an MSc (CGPA 5.00/5.00) from Bayero University Kano, and a First-Class BSc from BUK—where he was the best graduating student in Computer Science in both 2007, 2009, and 2011. He is ranked among the top 60 most-cited researchers worldwide on recommender systems according to Google Scholar, with

over 50 publications in Q1 journals including IEEE Access, PLOS One, Scientometrics, and Computers and Security.

His academic journey connects the consortium in two directions: outward, through an internationally recognised research career in data-driven systems and information retrieval; and inward, through deep roots at BUK, where he served successively as Graduate Assistant, Lecturer, and Deputy Dean of the Faculty of Computer Science and Information Technology. He is a former head of science at Kaduna State University and currently leads CIT at FUT Babura, where the centre's mandate spans internet services, ICT infrastructure, software development, and ICT consultancy for the university community.

What he brings to Cluster E:

- Recognised as a top-60 global researcher on recommender systems—algorithms with direct application to personalised safety tools, fraud detection, and misinformation filtering in digital platforms.
- Has published extensively on cybersecurity, mobile touch-screen authentication, malware detection, and digital trust in peer-reviewed IEEE and Scopus-indexed journals, including a systematic literature review on mobile authentication security (Computers and Security, Q1).
- Won multiple research grants: two TETFund IBR grants as Lead Researcher (totalling over ₦4.2 million), a Malaysian Government FRGS grant (MYR 120,000), and University of Malaya Research Grants—demonstrating a strong record of securing competitive funding.
- Brought Bronze Award at the International Invention and Innovative Competition (2017), Best Paper Presenter at PgRES 2017 (University of Malaya), and 1st Runner-Up in the University of Malaya 3-Minute Thesis Competition (2018).
- Research on context-aware systems, sentiment analysis of social media (including COVID-19 and public health discourse), and machine learning for student placement provides analytical tools applicable to online safety, consumer behaviour analysis, and digital literacy assessment.
- As Director of the CIT at FUT Babura, he brings institutional ICT infrastructure management experience, including internet services, university web platforms, and software development—practical capacity for supporting this programme's digital research infrastructure.
- Holds ICDL Africa certifications in Cybersecurity, Data Analytics, Digital Marketing, Collaboration, and Database Management, alongside professional memberships in the Internet Society (Nigeria Chapter) and the Nigerian Institute of Management.

4.5 Senior Research Specialist (Deep Learning and Computer Vision): Dr. Abdulrauf Garba Sharifai

Dr. Sharifai is a Senior Lecturer in the Department of Computer Science, Faculty of Computing, Northwest University, Kano, with 11 years of teaching and research experience. He holds a PhD in Computer Science from University Science Malaysia, an M.Eng from Liaoning University of Technology (China), and a BSc from Bayero University, Kano. He specialises in deep learning, feature selection, class imbalance, digital image processing, and computational intelligence.

What he brings to Cluster E:

- Research on feature selection for high-dimensional imbalanced biomedical datasets, published in Genes (MDPI) and IEEE Access—methods directly applicable to fraud detection, risk profiling, and anomaly detection in digital consumer environments [22].
- Deep learning research on COVID-19 auto-detection from X-ray images (Scientific Reports, 2024) and phishing attack detection using hybrid VGG19-TCN with multi-channel temporal attention—directly relevant to cybersecurity and consumer protection.
- Research on hybrid deep learning for network intrusion detection (CyberComp 2024, IEEE) and phishing detection using focal loss and self-attention CNNs, addressing online fraud and cybercrime.
- Research on text summarisation in African languages (presented 2025), with applications to automated content moderation and regulatory document analysis.
- Supervises two PhD candidates (semi-supervised object detection; multi-modal data retrieval) and four MSc candidates in deep learning and image processing.
- Reviewer for IEEE Access, IEEE Xplore, Elsevier, and PLOS One. Member of Malaysia Digital Economy Corporation (MDEC)-Intel AI Academic programme.
- BSc from Bayero University, Kano—ensuring institutional familiarity and smooth coordination with the Lead Institution.

4.6 Senior Research Specialist (Cyber Risk): Dr. Anas Abubakar Bisu

Dr. Bisu is Chief Cyber Risk Adviser at Cyber234 and Postdoctoral Research Associate at Durham University's Department of Engineering. He holds a PhD from Durham, an MEng from Portsmouth, and a First-Class BSc from BUK. Professional training from Oxford and the European Space Agency rounds out his credentials.

What he brings to Cluster E:

- Specialises in cyber risk management and secure communications infrastructure, areas that matter more as Nigeria's digital footprint grows.

- Researches 5G/6G network security and integrated satellite-terrestrial networks.
- Won an IEEE Best Track Paper Award, a UKRI-GCRF Grant, and an EU Horizon 2020 Marie Skłodowska-Curie Grant.
- Holds CCNA certification plus IEEE certifications in 5G, Blockchain, Edge Computing, IoT, and Risk Management.
- Worked as Network Engineer at Alcatel-Lucent in Nigeria and as Innovation Fellow at the Hellenic Telecoms Organization in Greece.
- Previously Senior Lecturer at Bayero University, Kano.

4.7 Senior Research Specialist (Cyber Security): Dr. Abubakar-Sadiq Shehu

Dr. Shehu is a Teaching Fellow in Cyber Security at the University of Leicester and a Lecturer in Information Technology at Bayero University, Kano—where he was pioneer staff in cybersecurity within the Faculty of Computer Science. He holds a PhD from the University of Porto (thesis: “Establishing Secure and Privacy-Preserving Digital Identity with Self-Sovereign Identity”), an MSc from the University of East London, and a BSc from the University of Wollongong, Australia. His career spans 12+ years across academia, big tech, banking, and international research programmes.

What he brings to Cluster E:

- Co-team researcher on NanoSTIMA (€2.5M EU Horizon 2020, INESC TEC Porto), building secure health data infrastructure used for Portugal’s COVID vaccine tracking system.
- Postdoctoral Research Fellow on DECaDE (£6M funded) at the Surrey Centre for Cyber Security. The programme’s work on secure digital economy in supply chains was presented to the House of Lords.
- Pioneer Language Engineer at Google (2022–2024): designed and optimised NLP algorithms and language models for Hausa and Yoruba, managing projects from pilot to production.
- Fraud Prevention Specialist at Apple (2020–2022): led fraud prevention across all payment channels, monitored debit card fraud alerts, identified fraud patterns by country and merchant, and reviewed VISA CAMS and FICO alerts.
- Relationship and Risk Officer at Unity Bank PLC (Nigeria, 2013–2015): conducted KYC/customer due diligence, developed cybersecurity strategies, and prepared risk reports for senior management—grounding his academic work in Nigerian banking realities.
- Founded SPIDVerify (spidverify.com): SaaS platform for identity verification, KYC compliance, business onboarding, and AML screening, now serving 500+

organisations globally.

- Publishes in IEEE EuroS&PW, Springer LNCS, and SECRIPT on decentralised identity, GDPR compliance, and SSI.
- Holds certifications in CCNA, Security+, CEH, GDPR Compliance, and ISO Security Standards. Fluent in Hausa, Yoruba, and English.

4.8 Senior Research Specialist (AI Lead): Prof. Salisu Mamman Abdulrahman

Professor Abdulrahman is a Professor of Artificial Intelligence (Machine Learning) and Dean of the Faculty of Computing and Mathematical Sciences at Kano University of Science and Technology (KUST), Wudil. He holds a PhD from the MAPi Joint Doctoral Programme in Computer Science (Universities of Minho, Aveiro, and Porto, Portugal—the same programme as the Consortium Lead), an MSc from Bayero University Kano, and a First-Class BSc from KUST Wudil. His Google Scholar profile records over 300 citations across machine learning, algorithm selection, and data-driven research.

What he brings to Cluster E:

- As Dean of the Faculty of Computing and Mathematical Sciences at KUST Wudil, he provides institutional leadership and governance capacity, supporting coordination between the two Kano-based Nigerian partner institutions.
- Research expertise in machine learning, algorithm selection, and meta-learning—published in *Machine Learning* (Springer, 2018) and the IEEE NIGERCON 2024 proceedings—provides the analytical foundations for automated detection of on-line harms, fraud, and deceptive practices.
- Research on low-resource language AI, including Hausa NLP and sentiment analysis, connects directly to the consortium’s content moderation and digital safety work [16].
- Extensive supervisory track record at MSc and PhD levels in AI and machine learning, strengthening the consortium’s doctoral cohort.
- Reviews for *Progress in Artificial Intelligence* (Springer) and *Machine Learning Journal* (Springer), and serves as external examiner for PhD and MSc programmes nationally (Nile University, UMYU, Federal University Dutse) and internationally (Bharathiar University, India).
- Registered with the Computer Professionals (Registration Council Nigeria), Reg. No. 006818/2020.

4.9 Senior Research Specialist (Policy and Governance): Prof. Bashir Shehu Galadanci

Prof. Galadanci holds a BSc (First Class) from King AbdulAziz University, and an MSc and PhD in Computer Engineering from Boston University, USA. He is a Professor of Software Engineering at BUK with 30+ years of academic, policy, and governance experience.

What he brings to Cluster E:

- Served as Special Advisor to the Governor of Kano State on ICT and Education, then as Commissioner for Science and Technology. He doesn't just study policy—he's made it.
- Currently National Coordinator of the UBEC Digital Resource Center, running digital education initiatives at a national scale.
- Held senior BUK positions: Dean of Faculty of Computer Sciences and IT, Head of Software Engineering, Senate member, Council member.
- Researches software engineering, technology adoption (UTAUT models), and comparative research output analysis across Nigerian universities.
- Co-authored NLP and machine translation work for African languages, contributing to the consortium's content moderation capabilities.

4.10 Professorial Lead: Prof. Muhammad Yahuza Bello

Prof. Yahuza Bello served as the 10th Vice Chancellor of Bayero University, Kano (2015–2020). He holds a PhD in Mathematics from the University of Arkansas and has 40+ years of academic leadership experience.

Why he matters:

- Founded Computer Science studies at BUK in 1990, which grew into a Faculty with over 1,500 graduates.
- Research in algebra and cryptography connects to the mathematical bedrock of cybersecurity and data protection.
- Served as Head of Mathematical Sciences, Dean of Science, Director of IT, Dean of Postgraduate Studies, and Deputy VC (Academics) before the vice-chancellorship.
- Supervised 6 PhD candidates, 37 MSc candidates, and over 50 BSc projects.
- Currently Pro-Chancellor of Al-Muhibbah Open University, Abuja.

4.11 Programme Administrator: Dr. Saminu Aliyu Kiri

Dr. Kiri, based at BUK, will coordinate research activities across all six institutions: managing documentation, tracking deliverables, supporting collaboration between Nigerian and international partners, and ensuring outputs are submitted on time. His familiarity with BUK’s research systems and the consortium’s institutional landscape positions him well for programme coordination, financial reporting, and stakeholder communication across the 36-month assignment.

5 Institutional Experience and Capacity

5.1 Nigerian Partner Institutions

5.1.1 Bayero University, Kano (Lead Institution)

BUK is one of Nigeria’s leading federal universities. Its Department of Computer Science and Faculty of Computer Science and Information Technology punch above their weight in NLP, AI, and digital safety research. BUK has managed grants from the Lacuna Fund, Google Research, and Oracle, and its research governance structures support multi-institutional collaboration. BUK is the institutional home of HausaNLP and Arewa Data Science, two of the most active NLP research communities on the continent.

5.1.2 Kano University of Science and Technology, Wudil

KUST Wudil, a state university in Kano, houses a Faculty of Computing and Mathematical Sciences with growing research output in AI, machine learning, and data science. Under the deanship of Prof. Abdulrahman, the faculty has developed strong doctoral and MSc programmes in AI and machine learning. KUST Wudil and BUK together form a natural Kano-based research corridor, sharing research networks, mentoring capacity, and access to state-level digital economy stakeholders. KUST’s First-Class graduates—including Prof. Abdulrahman himself—and its Computer Professionals (Registration Council Nigeria) membership underscore its professional standing.

5.1.3 Federal University of Technology, Babura, Jigawa State

The Federal University of Technology, Babura (FUT Babura) is a federal university in Jigawa State, northern Nigeria. Its Centre for Information Technology (CIT)—directed by Assoc. Prof. Haruna—provides internet services, ICT infrastructure management, software development, and ICT consultancy for the university community. FUT Babura extends the consortium’s geographic footprint from Kano into Jigawa State, widening its reach into underserved northern communities directly relevant to the Programme’s digital inclusion mandate.

Through Prof. Haruna's leadership, FUT Babura contributes a top-60 globally ranked researcher in recommender systems, a consistent record in cybersecurity publications (IEEE Access, Computers and Security), and competitive grant-winning experience from both Nigerian (TETFund) and international (Malaysian Government, University of Malaya) funding bodies. His decade-long academic foundation at BUK—where he was best graduating student and served as Lecturer and Deputy Dean—ensures effective institutional coordination with the Lead Institution.

5.1.4 Northwest University, Kano

Northwest University, Kano is a Nigerian state university with a Faculty of Computing. Through Dr. Sharifai, the consortium gains expertise in deep learning, computer vision, feature selection, and cybersecurity. His research on network intrusion detection, phishing detection, and image-based anomaly detection—all published at IEEE venues—adds a complementary technical layer to the consortium's digital safety and consumer protection work. His BSc from BUK and 11 years of continuous teaching experience in northern Nigeria ensure effective coordination with the Lead Institution and access to the same regional research networks.

5.2 International Partner Institutions

5.2.1 York St John University, UK

York St John hosts the Cybersecurity Research Group directed by Dr. Usman. Active programmes in trust-based systems, data privacy, biometric security, IoT, and the Cyber360 education platform add UK cybersecurity network access and international research standards. The group's doctoral supervision record in biometric authentication and privacy-by-design frameworks directly adds to the consortium's research capacity.

5.2.2 University of Leicester, UK

The University of Leicester's School of Computing and Mathematical Sciences runs active programmes in cybersecurity, information security, and digital forensics. Through Dr. Shehu—who also holds a lectureship at BUK—Leicester contributes expertise in self-sovereign identity, GDPR-compliant privacy systems, fraud prevention, and forensic investigation. His prior appointments at the Surrey Centre for Cyber Security (DECaDE, £6M) and INESC TEC Porto (NanoSTIMA, €2.5M EU Horizon 2020) give the consortium access to established European cybersecurity research networks. His dual Nigerian-UK appointment bridges international research standards and Nigerian institutional needs directly.

6 Doctoral and Post-Doctoral Research Capacity

One of this Programme’s most valuable long-term outcomes won’t be a report or a dataset. It will be people. A cohort of researchers who can sustain top-tier work in digital trust and safety long after the funding ends. We take that seriously—and we don’t say it in the abstract. We’ve already built the infrastructure to prove it.

6.1 A Proven Mentoring Track Record

The Consortium Lead, Dr. Muhammad, co-founded two community-driven initiatives that have trained hundreds of Nigerian students and early-career researchers:

- **Arewa Data Science Academy** (arewadatascience.github.io) is a free, 16-week fellowship in data science, machine learning, and deep learning. Backed by Google Research, TensorFlow, and BUK, the Academy has graduated multiple cohorts—three in machine learning, two in deep learning—with a deliberate focus on underrepresented groups including women and communities affected by insecurity. Over a dozen mentors from BUK, FUT Babura, and international institutions—including Dr. Ahmad, Dr. Abdulmumin, and other consortium members—guide fellows through structured courses (30 Days of Python, Machine Learning, Deep Learning, Kaggle Challenges) and capstone research projects.
- **HausaNLP** (hausanlp.github.io) is an open-source community for Hausa NLP, co-founded by Dr. Muhammad and Dr. Abdulmumin. The community has produced datasets, word embeddings, machine translation systems, and sentiment benchmarks now used by researchers worldwide. Its curated catalogue (catalog.hausanlp.org) aggregates all known Hausa NLP resources. HausaNLP collaborates with Masakhane, Black in AI, and the Open Language Data Initiative.

These aren’t side projects. They are the mentoring infrastructure that this programme will draw upon. Fellows and mentors from these communities represent a ready pool of junior and mid-career researchers who can contribute to Cluster E from the outset—annotating datasets, running experiments, conducting fieldwork, and eventually producing their own doctoral research on digital trust and safety topics.

6.2 Meeting the Programme’s Researcher Requirements

We commit to recruiting at least fifteen postdoctoral researchers and fifteen PhD candidates. The goal is researchers who can bridge technical skill and policy impact—people who produce evidence *and* know how to get it in front of decision-makers.

- **Postdoctoral researchers:** Community networks (Masakhane, HausaNLP, Arewa Data Science, Black in AI) and institutional affiliations connect us to a broad postdoctoral talent pool. Several Arewa Data Science alumni and HausaNLP

contributors have since completed or are completing doctoral degrees, creating a natural feeder into postdoctoral positions.

- **PhD candidates:** BUK, FUT Babura, KUST Wudil, and our international partners run active doctoral programmes. We will recruit PhD candidates whose dissertations address nationally relevant Cluster E questions. The Arewa Data Science fellowship model provides a tested approach to identifying promising candidates from underrepresented backgrounds.
- **Mentoring structure:** Our senior researchers have supervised dozens of doctoral and postdoctoral researchers through formal academic channels. The Arewa Data Science model adds a structured, cohort-based approach: dedicated mentors, regular workshops, and peer networks that complement traditional academic supervision. We will apply this model to Cluster E’s postdoctoral and PhD cohorts.
- **Professorial Lead:** Professor Yahuza Bello (former BUK Vice Chancellor) provides academic leadership and quality assurance. His governance experience and institutional networks give the programme the stature it needs.
- **Reach beyond the consortium:** Through HausaNLP, Arewa Data Science, AfriSafe, and Masakhane, the programme’s impact won’t stay locked inside six institutions. These open networks pull in emerging researchers across Nigeria and the continent. When this programme ends in 36 months, the community infrastructure will still be running. That’s by design.

7 Existing Platforms and Research Infrastructure

We don’t start from scratch. The consortium already operates platforms and projects that give Cluster E a running start.

7.1 AfriSafe: Online Safety in the Age of AI

AfriSafe (<https://afrisafe.vercel.app>) addresses a problem growing more urgent every month: safety in an era of generative AI. The project builds resources, tools, and frameworks for understanding AI-related harms—deepfakes, AI-generated misinformation, automated hate speech. With programme funding, AfriSafe will sharpen its focus on Nigerian content, developing safety resources built for Nigeria’s linguistic diversity and digital landscape. Research Streams 1 and 5 gain immediate infrastructure from day one.

7.2 VigilSentinel: Cybersecurity Operations Platform

VigilSentinel (www.vigilsentinel.com), founded by Dr. Usman, consolidates vulnerability assessment, endpoint monitoring, threat detection, log collection, and reporting in a single web application. Over 15 assessment types. Self-hosted or cloud-hosted.

Role-based access control. For this programme, it serves as a real-world testbed for evaluating digital safety interventions under Research Stream 3.

7.3 Wazobia Invoice: Secure FinTech for Underserved Markets

Wazobia Invoice (www.wazobia-invoice.ng), also Dr. Usman's creation, helps small Nigerian businesses manage invoices and financial records securely—even offline. Its privacy-preserving, offline-first architecture makes it a working tool for financial inclusion in low-connectivity areas. For Research Streams 2 and 3, it offers a case study in consumer protection and trust in digital financial systems.

7.4 SPIDVerify: Identity Verification and KYC Compliance

SPIDVerify (spidverify.com), founded by Dr. Shehu, provides SaaS-based identity verification, KYC compliance, business onboarding, and AML screening, now serving 500+ organisations globally. It connects Dr. Shehu's academic work on self-sovereign identity to a live commercial product. For Research Streams 2 and 4, SPIDVerify serves as both a case study in identity-based consumer protection and a testbed for evaluating how KYC and AML tools can be adapted to Nigerian regulatory and market conditions.

Together with our datasets (AfriHate, AfriSenti, NaijaSenti, HERDPhobia) and community networks (HausaNLP, Arewa Data Science, Masakhane), these four operational platforms give the consortium research tools that most teams would spend their first year building. VigilSentinel, Wazobia Invoice, SPIDVerify, and AfriSafe—all built by consortium members—are already deployed in real environments. That reflects a team that treats the gap between academic insight and practical application as something to close, not accept.

8 Indicative Research Programme

Full research questions and methodology will follow in the Technical Proposal. The consortium proposes six interconnected research streams.

8.1 Research Stream 1: Online Harms, Hate Speech, and Content Moderation

This stream addresses detection, governance, and mitigation of harmful online content in Nigeria's digital spaces.

- Build multilingual hate speech and abusive language detection systems for Nigerian languages, extending AfriHate and NaijaSenti.
- Evaluate how well content moderation by major platforms fits Nigeria's cultural and linguistic realities.

- Design community-informed governance models for content moderation that respect Nigeria’s diversity.
- Study how political polarisation, misinformation, and radicalisation play out in Nigerian digital spaces.

Findings will inform national policies on online safety, platform governance, and content moderation—providing evidence for context-aware regulatory responses to online harms.

8.2 Research Stream 2: Consumer Protection in Digital Markets

This stream addresses online fraud, deceptive practices, and consumer exploitation in Nigeria’s digital and financial ecosystems.

- Map the landscape of online fraud, digital financial harms, and consumer exploitation in Nigeria’s growing digital economy.
- Develop NLP tools for detecting fraudulent content, phishing, and deceptive practices in Nigerian languages.
- Assess whether existing consumer protection mechanisms and regulatory frameworks work in practice.
- Study how women, youth, and rural communities experience digital marketplaces—and where they are most at risk.

Findings will inform national consumer protection, digital financial regulation, and online safety policies.

8.3 Research Stream 3: Secure and Privacy-Preserving Authentication for Trustworthy FinTech

This stream addresses the security and trust foundations required for Nigeria’s rapidly expanding digital financial sector.

- Design and evaluate multi-modal continuous authentication frameworks combining behavioural biometrics, device signals, and contextual indicators, matched to Nigerian FinTech usage patterns.
- Address challenges faced by underserved groups—rural users, informal workers, women, and persons with disabilities—by reducing reliance on static credentials such as passwords and one-time PINs.
- Produce policy guidance on balancing fraud prevention, consumer protection, and data privacy in digital financial platforms.

Cluster alignment: Cluster E (Trust, Safety, Consumer Protection), Cluster B (Digital Public Infrastructure and Digital Government).

8.4 Research Stream 4: Trust, Privacy, and Data Protection Frameworks

- Evaluate how well Nigeria’s data protection framework (NDPA 2023 and NDPR) is implemented and enforced. Recommend improvements.
- Develop privacy-preserving technical frameworks for digital identity, health systems, and fintech.
- Investigate public trust in digital platforms and government services, with attention to cultural and demographic variation.
- Compare data protection enforcement models across African and international jurisdictions.

8.5 Research Stream 5: Youth Online Safety and Digital Literacy

- Assess online harms affecting Nigerian youth: cyberbullying, harmful content exposure, online exploitation.
- Evaluate existing digital literacy programmes and propose evidence-based improvements.
- Develop age-appropriate safety tools and resources in Nigerian languages.

8.6 Research Stream 6: Regulatory Innovation and Adaptive Governance

- Compare digital trust and safety regulatory frameworks across the EU, AU, ECOWAS, UK, and India.
- Design adaptive governance mechanisms that promote innovation without abandoning citizen protection.
- Assess whether Nigerian regulatory agencies have the institutional capacity to enforce digital safety standards.
- Recommend reforms and capacity-building strategies that make regulation more effective.

9 Policy Outreach and Impact Pathways

Research that doesn’t reach decision-makers is an expensive hobby. Our outreach strategy is built to prevent that.

- **Target MDAs:** FMCIDE, Nigeria Data Protection Commission (NDPC), NITDA, NCC, National Orientation Agency, Federal Ministry of Education, and relevant state-level agencies.

- **Policy instruments:** Regular policy briefs, executive summaries, and presentations—each shaped for specific stakeholders and timed to land at decision points.
- **Stakeholder dialogue:** Structured conversations with government, tech platforms, fintech companies, civil society, and development partners.
- **Capacity building:** Workshops and training for government officials on digital trust, online safety, and evidence-based policy-making.
- **Open data:** All datasets and outputs delivered in reusable, machine-readable formats. If other researchers or agencies can't build on our work, we haven't finished the job.

10 Eligibility Statement

All participating institutions satisfy the World Bank's *Procurement Regulations for IPF Borrowers* (February 2025) regarding Conflict of Interest (paragraphs 3.15, 3.16), Unfair Competitive Advantage (paragraph 3.17), and Eligibility (paragraphs 3.20–3.23). Supporting documentation accompanies this EOI.

Bayero University, Kano—the lead institution—is a Nigerian university eligible under the World Bank's *Procurement Regulations*. All consortium members are accredited academic and research institutions with no known disqualifying conflicts.

11 Contribution to Project BRIDGE Objectives

Project BRIDGE is deploying at least 90,000 km of fibre-optic backbone. That will bring millions of Nigerians online—many for the first time. But infrastructure without knowledge is a road that leads to risk. Our research ensures connectivity and governance travel together.

- **Inclusive connectivity:** New users need safety and trust mechanisms from day one, not as an afterthought. Our work makes sure expanded connectivity protects all users, especially underserved communities.
- **Digital inclusion:** Trust barriers keep women, youth, persons with disabilities, and rural communities from fully joining the digital economy. We address those barriers with culturally grounded, linguistically diverse safety tools.
- **Economic transformation:** Digital commerce, fintech, and e-government need strong consumer protection. Our real-world platforms (Wazobia Invoice, VigilSentinel, SPIDVerify) prove that secure, locally designed digital solutions work.
- **National research capacity:** We're building a pathway from PhD candidate to postdoc to established professor in digital trust and safety. The goal: Nigeria producing the research that shapes its own digital future, not importing it. Our

community networks extend this beyond formal academic institutions.

- **A replicable model:** This cluster combines university research, industry platforms, and government policy experience into one coordinated effort. Done well, it shows how future national research programmes can move beyond papers to real institutional change.

12 Conclusion

This consortium carries a record in exactly the areas Cluster E demands: hate speech detection, online safety, cybersecurity, trust systems, NLP for African languages, data privacy, AI governance, and digital identity. We've published at top venues, won major grants, built working platforms, and founded research communities. Those credentials are the entry ticket.

But the real test won't be measured in publications. It will show up in better policy. Stronger institutions. A more capable generation of Nigerian researchers. Research that turns into governance tools and safety mechanisms Nigerians can actually use.

That's what we're here to build. Academic research, industry platforms, and government policy experience—brought together into one research programme. Our aim: research findings that don't gather dust in journals but become the frameworks, regulations, and capabilities Nigeria needs as it builds one of Africa's most ambitious digital infrastructure programmes.

The countries that will lead in the decades ahead aren't just those that lay the most fibre. They're the ones that organise themselves to make connectivity safe, trusted, and inclusive. We're ready to help Nigeria be one of them.

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Position	Name	Affiliation	Qualification	Experience
Consortium Lead	Dr. Shamsuddeen Hassan Muhammad	BUK / Imperial College London	PhD Computer Science	15+ yrs; NLP, AI, online harms
Senior Research Specialist	Dr. Aminu Bello Usman	York St John University, UK	PhD Cyber Security	15+ yrs; cybersecurity, trust, privacy, fintech
Senior Research Specialist	Dr. Idris Abdulmumin	Univ. of Pretoria / ABU Zaria	PhD Computer Science	7+ yrs; NLP, hate speech, sentiment
Senior Research Specialist (NLP & Ethical AI)	Dr. Ibrahim Said Ahmad	Bayero University, Kano	PhD Computer Science	10+ yrs; NLP, ethical AI, emotion detection, LLMs
Senior Research Specialist (Data & Recommender Systems)	Assoc. Prof. Khalid Haruna	FUT Babura / BUK	PhD Computer Science	10+ yrs; recommender systems, data analytics, cybersecurity
Senior Research Specialist (Deep Learning & CV)	Dr. Abdulrauf Garba Sharifai	Northwest Univ., Kano	PhD Computer Science	11+ yrs; deep learning, feature selection, computer vision, cybersecurity
Senior Research Specialist (Cyber Risk)	Dr. Anas Abubakar Bisu	Durham University / BUK	PhD Engineering	10+ yrs; cyber risk, network security, 5G/6G
Senior Research Specialist (Cyber Security)	Dr. Abubakar-Sadiq Shehu	Univ. of Leicester / BUK	PhD Computer Science	12+ yrs; SSI, fraud prevention, GDPR, forensics
Senior Research Specialist (AI Lead)	Prof. Salisu Mamman Abdulrahman	KUST Wudil	PhD Computer Science	15+ yrs; AI, ML, data-driven systems
Senior Research Specialist (Policy & Governance)	Prof. Bashir Shehu Galadanci	Bayero University, Kano	PhD Computer Engineering	30+ yrs; ICT policy, governance, software engineering
Professorial Lead	Prof. Muhammad Yahuza Bello	Bayero University, Kano	PhD Mathematics	40+ yrs; academic leadership, IT, cryptography
Data & Analytics Specialist	[To be confirmed]	[TBD]	[TBD]	5+ yrs data analysis
Programme Administrator	Dr. Saminu Aliyu Kiri	Bayero University, Kano	PhD	Research administration and programme coordination